

1.1 Background

Large language models (LLMs) have emerged as one of the most significant advances in artificial intelligence, demonstrating remarkable capabilities in language understanding, text generation, and multi-step reasoning (*Brown et al., 2020; OpenAI, 2024*). Models such as the GPT series (*Brown et al., 2020; OpenAI, 2024*) and the Llama family (*Touvron et al., 2023; Grattafiori et al., 2024*) achieve strong performance across diverse tasks by learning statistical patterns from massive text corpora. This extensive training enables them to generate coherent responses across a wide range of domains without requiring the explicit storage of factual knowledge in a structured form.

Despite these capabilities, LLMs remain susceptible to hallucination, the generation of fluent but factually incorrect or unverifiable content (*Ji et al., 2023*). Huang et al. (2023b) distinguish between intrinsic hallucinations, where generated content contradicts the provided source, and extrinsic hallucinations, where the content cannot be verified using any reliable source. Hallucination is particularly problematic in knowledge-intensive applications because autoregressive language models generate each token based on previously generated tokens rather than by verifying facts against an external knowledge source. Consequently, an incorrect reasoning step may propagate through the remainder of the generation process, producing outputs that appear convincing despite being factually inaccurate (*Ji et al., 2023; Huang et al., 2023b*). Recent theoretical work further suggests that hallucination is an inherent limitation of statistical language modelling rather than merely a consequence of insufficient training, implying that improved scaling or prompting alone cannot completely eliminate it (*Xu et al., 2025*).

Knowledge Graph Question Answering (KGQA) represents one of the most demanding applications of factual reasoning. A knowledge graph (KG) is a structured collection of verified facts represented as (entity, relation, entity) triplets, for instance, (Paris, capital of,

France), enabling reasoning through interconnected relationships (*Bollacker et al., 2008*).

Standard KGQA benchmarks such as WebQSP (*Yih et al., 2016a*) and ComplexWebQuestions (*Talmor and Berant, 2018a*) are built on the Freebase knowledge graph and frequently require multi-hop reasoning across several connected entities. Because such questions depend on precise factual relationships, unconstrained LLMs relying solely on parametric memory can easily generate plausible but incorrect reasoning paths (*Lan et al., 2022a; He et al., 2021a*).

Retrieval-Augmented Generation (RAG) has become the dominant strategy for improving factual grounding by retrieving relevant external information and supplying it to the LLM during inference (*Lewis et al., 2020b; Izacard and Grave, 2021*). Although RAG significantly reduces hallucination, the retrieved information only influences generation rather than constraining it. Consequently, the model may still ignore relevant evidence or generate unsupported claims beyond the retrieved context (*Asai et al., 2024; Agrawal et al., 2024b*).

Knowledge graph-constrained decoding addresses this limitation by restricting the model to generate only tokens that correspond to valid paths within a knowledge graph. During decoding, a constraint oracle applies a validity mask over the model's output vocabulary, preventing the selection of tokens that violate the graph structure (*Geng et al., 2023; Willard and Louf, 2023*). Unlike RAG, which provides soft guidance, constrained decoding imposes hard constraints that guarantee every generated reasoning step corresponds to verified knowledge graph facts.

Graph-Constrained Reasoning (GCR) (*Luo et al., 2025a*) is the leading implementation of this paradigm. GCR constructs a pre-computed KG-Trie that serves as the constraint oracle during beam-search decoding, ensuring that every generated reasoning chain follows a valid knowledge graph path and thereby achieving complete structural faithfulness. However, because the trie is constructed before decoding begins, it remains static throughout the generation process. It does not consider the semantic intent of the input question or the

evolving reasoning context represented by the partially generated output. As a result, the trie often contains many structurally valid but semantically irrelevant paths, requiring the LLM to rely on its internal parametric knowledge to distinguish relevant candidates, precisely the behaviour constrained decoding seeks to minimise (Li et al., 2025; Luo et al., 2025a).

Decoding on Graphs (DoG) (Li et al., 2025) improves upon GCR by dynamically expanding the constraint trie during decoding to reflect the current reasoning direction. However, the framework still admits all structurally valid neighbouring nodes regardless of their semantic relevance to the question or partial reasoning chain. Consequently, the model must continue to discriminate between relevant and irrelevant candidates using parametric knowledge while incurring the additional computational cost of rebuilding the constraint structure during inference.

These limitations motivate the Dynamic Context-Aware Trie (DCA-Trie) framework proposed in this project. DCA-Trie extends existing KG-constrained decoding by incorporating semantic relevance directly into the constraint oracle, conditioning the valid token set on the knowledge graph structure, the input question, and the partially generated reasoning chain. By combining structural constraints with dynamic semantic filtering, the proposed framework aims to reduce constraint permissiveness, improve answer accuracy, and maintain the structural faithfulness guarantees provided by existing constrained decoding methods.

1.1 Problem Statement

Existing knowledge graph-constrained decoding frameworks, particularly Graph-Constrained Reasoning (GCR) (Luo et al., 2025a) and Decoding on Graphs (DoG) (Li et al., 2025), construct their constraint oracles as functions $f(G, E_q)$ of the knowledge graph and question entities. This design successfully guarantees structural faithfulness by ensuring that every generated token corresponds to a valid knowledge graph path. However, their constraint

oracles are constructed primarily from the knowledge graph structure and the identified question entities, without adequately incorporating the semantic intent of the input question or the evolving context of the partially generated reasoning chain (*Geng et al., 2023; Willard and Louf, 2023*).

As a result, the valid token set often contains numerous structurally valid but semantically irrelevant paths. Although these paths satisfy graph constraints, they are unrelated to the user's information need, forcing the LLM to rely on its internal parametric knowledge to identify the most appropriate reasoning path. This residual dependence on parametric knowledge weakens the effectiveness of constrained decoding and becomes increasingly problematic in multi-hop knowledge graph question answering, where the number of candidate paths grows rapidly with reasoning depth (*Lan et al., 2022a; He et al., 2021a; Talmor and Berant, 2018a*).

Recent work such as RouterKGQA (*Yuan et al., 2026*) introduces semantic filtering to improve efficiency and answer quality. However, semantic filtering is applied after generation rather than during decoding. Consequently, the language model remains exposed to semantically irrelevant candidates throughout the generation process, thereby limiting its ability to eliminate unnecessary reasoning and preserve deterministic constraint-based generation.

This project addresses this limitation by proposing the Dynamic Context-Aware Trie (DCA-Trie), which extends the constraint oracle from $W_{\text{val}}(t) = f(G, E_q)$ to $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$. The proposed framework conditions the valid token set on the knowledge graph structure, the input question, and the partially generated reasoning chain, enabling semantic filtering throughout the decoding process while preserving structural faithfulness.

The effectiveness of the proposed framework will be evaluated using three criteria: (i) preservation of structural faithfulness, ensuring that every generated token corresponds to a valid knowledge graph path; (ii) reduction in semantic irrelevance, measured using the

Semantic Irrelevance Ratio (SIR) introduced in this study; and (iii) maintenance of answer recall by ensuring that semantic filtering does not remove valid gold answer paths during inference.

1.3 Aim and Objectives

Aim

This project aims to design, implement, and evaluate a Dynamic Context-Aware Trie (DCA-Trie) framework for knowledge graph-constrained large language model generation. The framework conditions the valid token constraint at each decoding step on the knowledge graph structure, the input question, and the partially generated reasoning chain to improve answer accuracy in multi-hop knowledge graph question answering while maintaining 100% structural faithfulness.

Objectives

To achieve this aim, the following objectives have been established:

- i. To characterize the constraint permissiveness problem in existing KG-constrained decoding frameworks by defining and measuring the Semantic Irrelevance Ratio (SIR) of GCR’s static KG-Tries on the WebQSP and CWQ benchmarks across different reasoning depths.
- ii. To design a semantic relevance scoring mechanism using pretrained sentence transformer embeddings (*Reimers and Gurevych, 2019a*) for filtering candidate knowledge graph paths according to their relevance to the input question and the evolving reasoning context without task-specific fine-tuning.
- iii. To implement a static semantic filtering framework (DCA-Trie v1) that applies relevance scoring during KG-Trie construction, reducing trie size while maintaining a false negative rate below 5% for gold answer paths on the WebQSP validation set.
- iv. To implement a step-wise dynamic expansion variant (DCA-Trie v2) in which the trie is

updated after each committed triplet using the relevance scorer conditioned jointly on the question and the partial generation $y_{<t}$, directly operationalizing the

$W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$ formulation.

v. To evaluate DCA-Trie v1 and DCA-Trie v2 against the GCR baseline and an unconstrained Chain-of-Thought (CoT) baseline on the WebQSP and CWQ benchmarks using Hits@1, F1-score, structural faithfulness, Semantic Irrelevance Ratio (SIR), and average trie size, with results analysed according to question hop depth.

vi. To demonstrate the practical applicability of DCA-Trie by developing an experimental prototype that enables users to submit natural language questions and obtain factually grounded reasoning chains and answers generated through the proposed framework.

1.4 Tools and Facilities Used

1.4.1 Software Tools

Table 1.1 presents an evaluation of the software tools implemented in this project, their role within the DCA-Trie framework, and the rationale for their selection.

Table 1.1 Implemented Software Tools Evaluation

Tool	Use in Project	Rationale
Python / HuggingFace Transformers	Model inference pipeline; loading and running GCR’s Llama-3.1-8B checkpoint; integrating the relevance scorer	Provides reliable model loading, tokenization, and generation utilities, and enables direct use of the released GCR checkpoint.
GCR Framework	Primary baseline; KG-Trie construction, graph-constrained decoding, and result reproduction on WebQSP and CWQ	Provides a strong baseline and supports the trie-based constrained decoding mechanism that DCA-Trie extends.
Sentence- transformers / all-MiniLM-L6-v2	Semantic relevance scoring; computing cosine similarity between question-context and candidate KG path embeddings	Provides fast semantic embeddings without extra fine-tuning and supports efficient iterative decoding.
PyTorch	Deep learning framework underpinning all model inference and custom decoding logic	Supports implementation and testing of custom decoding logic and integrates smoothly with HuggingFace.
Freebase / WebQSP and CWQ Datasets	Knowledge graph source and KGQA benchmarks for evaluation	Provides standard datasets for this research area and enables direct comparison with prior work.

Google Colab Pro (A100, 40 GB)	GPU environment for all inference experiments; Llama-3.1-8B constrained decoding with beam search	Provides enough GPU memory for stable experiments and removes the need for local high-end hardware.
Visual Studio Code + Git / GitHub	Development environment and version control for all four team members	Supports day-to-day coding, team collaboration, version tracking, and reproducibility throughout the project.

1.4.2 Facilities

The following facilities were utilized in the execution of this project:

1.4.2.1 Google Colab Pro (A100, 40 GB VRAM) for all model inference and constrained decoding experiments.

1.4.2.2 University library and online academic databases (ACM Digital Library, arXiv, Google Scholar, ACL Anthology) for literature review and citation verification.

1.4.2.3 University internet infrastructure for cloud connectivity, dataset access, and retrieval of model checkpoints from the HuggingFace model hub.

1.4.2.4 Shared team repository hosted on GitHub for version control, experiment logging, and reproducibility of all reported results.

1.5 Methods Used

The project employs the following methods:

- i. Systematic review and critical analysis of the academic literature on constrained decoding, knowledge graph question answering, hallucination in LLMs, and retrieval-augmented generation.
- ii. Formal gap analysis of the four foundational papers, GCR (Luo *et al.*, 2025a), DoG (Li *et al.*, 2025), GCD (Geng *et al.*, 2023), and Outlines (Willard and Louf, 2023), to

identify the specific limitations that motivate the DCA-Trie design, including the permissiveness problem and the absence of generation-state feedback in the constraint oracle.

- iii. Formal specification of the DCA-Trie framework, including definition of the Semantic Irrelevance Ratio metric, the relevance scoring mechanism using sentence transformer embeddings (Reimers and Gurevych, 2019a), the static filtering procedure for v1, and the step-wise dynamic expansion rule for v2.
- iv. Reproduction of the GCR baseline on the WebQSP benchmark using GCR’s publicly released Llama-3.1-8B checkpoint to establish a verified starting point within 1–2% of published Hits@1.
- v. Implementation of DCA-Trie v1 and v2 as extensions to GCR’s inference pipeline, with relevance threshold selected on a 100-question held-out validation subset of WebQSP to minimize false negative rate on gold answer paths.
- vi. Quantitative evaluation of all system configurations on WebQSP and CWQ using Hits@1, F1, faithfulness rate, SIR, and average trie size per step, stratified by question hop depth following the evaluation protocol of Lan *et al.* (2022a).
- vii. Ablation studies examining the sensitivity of DCA-Trie’s performance to the relevance threshold τ , the sentence encoder architecture, and beam size k , isolating the contribution of each mechanism to the overall result.
- viii. Qualitative error analysis identifying failure cases in which DCA-Trie’s semantic filter erroneously excludes a gold answer path, supporting honest characterization of the

framework’s limitations and providing concrete directions for future work.

1.6 Scope of Work

This project designs and evaluates a Dynamic Context-Aware Trie framework for KG-constrained LLM decoding, with the specific aim of addressing the permissiveness problem identified in GCR and DoG. The framework is developed and evaluated in the context of multi-hop KGQA over Freebase-grounded benchmarks (WebQSP and CWQ), which are the standard evaluation settings in the KG-constrained decoding literature. All experiments use GCR’s publicly released Llama-3.1-8B checkpoint; no new model training or fine-tuning is performed. The semantic relevance scorer uses the all-MiniLM-L6-v2 sentence transformer without domain-specific adaptation. We emphasize that all experiments use GCR’s publicly released Llama-3.1-8B checkpoint (Luo *et al.*, 2025a), which includes their fine-tuned KG-specialized parameters. No additional model training or fine-tuning is performed in this work; our contribution is strictly the constraint oracle architecture and semantic filtering mechanism.

The scope is explicitly bounded as follows. Generalization of DCA-Trie to knowledge graphs other than Freebase, or to non-KGQA tasks such as entity linking or structured information

extraction, is identified as future work and is not evaluated in this dissertation. Formal proofs of the faithfulness guarantee under dynamic pruning are beyond scope; empirical measurement of the gold answer false negative rate is provided as evidence. The confidence-conditioned constraint tightening mechanism, in which constraint granularity is modulated by the LLM’s token-level entropy, is described as a theoretical extension but is not implemented. Production-scale deployment, latency optimization for online serving, and integration with proprietary enterprise knowledge graphs are engineering extensions beyond the research scope of this project.

The primary contribution of this work is the DCA-Trie framework and the Semantic Irrelevance Ratio metric. Secondary contributions are the empirical characterization of constraint permissiveness in existing frameworks and the ablation analysis isolating the contribution of semantic filtering from dynamic expansion. All code, experimental configurations, and the curated evaluation datasets will be made publicly available to support reproducibility and future extension.

1.7 Organization of Work

Chapter 1 introduces the study by identifying the research problem, stating the project aim and objectives, defining the proposed DCA-Trie framework, and outlining the scope and organization of the project. Chapter 2 presents the literature review by examining existing research on hallucination in large language models (LLMs), Knowledge Graph Question Answering (KGQA), constrained decoding, and related frameworks, highlighting the limitations that motivate the development of a context-aware constraint mechanism. Chapter 3 presents the problem formulation and research methodology, including the upgraded constraint framework, research questions, evaluation criteria,

and the implementation of DCA-Trie v1 and DCA-Trie v2 within the Graph-Constrained Reasoning (GCR) pipeline (Luo et al., 2025a). Chapter 4 presents the results and discussion by reporting the experimental evaluation on the WebQSP and CWQ benchmarks, analysing structural faithfulness, semantic irrelevance, and answer accuracy, and discussing the findings through ablation and error analysis. Chapter 5 concludes the dissertation by summarising the key findings and contributions of the study, discussing its limitations, and providing recommendations for future research.

